



Fig. 3: IPS and TN comparison of the first image

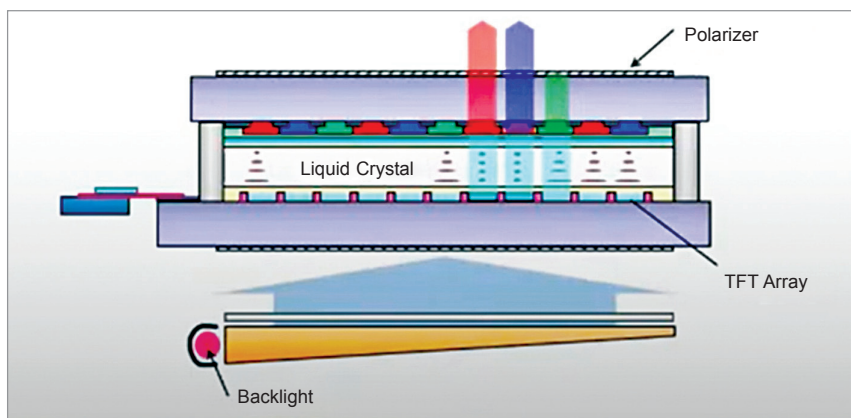


Fig. 4: Thin film transistor liquid crystal display (TFT LCD)



Fig. 5: Memory LCD applications

tom is the backlight.

The open cell has a nickel crystal sandwiched between the colour filter glass and the TFT array glass on which the transistor is built. The backlight structure contains a lot of optical sheets like diffuser, prisms,

light guide, and reflectors. Backlight is of two types, namely, direct light that refers to the LEDs which are directly laid below the LCD. This method is used for big-size LCDs like TV or signage. For small- and medium-size LCDs, like phones or

laptops, the side light is used where the LEDs are to the side of the LCDs.

Memory LCD for IoT displays

Memory LCD is the integration of one-bit memory into each LCD pixel. This makes sure that data transmission from outside the module is not needed when the displayed image does not change. The advantages that they offer are:

Low energy and power consumption. Two main reasons exist for the same. First the reflective displays do not need a backlight. Second, the memory needed integrated with the pixel does not need to be refreshed when showing a statistical image. For instance, in a 6.9cm (2.7-inch) screen, the power consumed is only $50\mu\text{W}$ when there is no image updating, whereas when an image data updates the power consumed is $175\mu\text{W}$. The important thing to note here is that, unlike conventional LCDs where power is consumed at milliwatt (mW) level, TFT LCDs consume power at a microwatt (μW) level, confirming the long life of the battery.

Stylish display quality. The high resolution and crisp graphic image help for both indoor and outdoor readability. The high reflectivity makes it feel as though a paper is being read.

Engineer-friendly. The design interface is very simple with a 3-wire serial signal input (clock, data, and chip select), along with a ground and power supply.

Since the location of the sensors in IoT must be remote and needs a longer battery life, memory LCDs are widely recommended. Fig. 5 shows applications where memory LCDs can be majorly used, which are available from 2.5cm (1 inch) to 11.2cm (4.4 inch), in both black-and-white and colour. They can also be used in smart watches.

IGZO LCD for smartphones

The materials used to build the TFT LCD are indium, gallium, zinc, and oxygen (IGZO).